Controllability of Prabhakar Fractional Dynamical Systems

Md Samshad Hussain Ansari, Muslim Malik, and Dumitru Baleanu

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Abstract

This paper investigates the exact controllability of nonlinear fractional dynamical systems involving the Prabhakar fractional derivative with Mittag-Leffler kernels. Using fractional calculus, semigroup theory, and Banach fixed-point theorems, sufficient conditions for exact controllability are established. An illustrative example is presented to validate theoretical results.

1. Introduction

Fractional calculus has emerged as an indispensable mathematical tool for modeling anomalous diffusion, memory effects, and hereditary properties in complex physical and engineering systems. In particular, Prabhakar fractional calculus involves three-parameter Mittag-Leffler functions in its kernel, generalizing Riemann-Liouville and Caputo fractional derivatives.

Controllability is a fundamental qualitative concept in mathematical control theory. A dynamical system is called controllable on a finite interval [0, T] if an admissible control input steering the system state from an arbitrary initial state to an arbitrary final state exists.

2. Preliminaries and System Formulation

Let X and U be real Banach spaces representing the state and control spaces, respectively. We denote by C([0, T], X) the space of continuous functions with supremum norm.

> ***Definition 2.*** *1 (Prabhakar Fractional Derivative). Let alpha > 0, beta > 0, gamma in R, and omega in R. The Prabhakar fractional derivative of a function f(t) is defined by D^{alpha, beta, gamma}_{omega} f(t) = (d/dt) int_0^t (t - s)^{-beta} E_{alpha, 1-beta}^{-gamma}(omega (t - s)^alpha) f(s) ds, where E_{alpha, beta}^gamma(z) denotes the three-parameter Mittag-Leffler function.*

Consider the following nonlinear Prabhakar fractional dynamical control system:

`D^{alpha, beta, gamma}_{omega} x(t) = A x(t) + B u(t) + f(t, x(t)), t in [0, T], x(0) = x_0`

where x(t) in X is the state, u(t) in U is the control input, A: D(A) subset X -> X generates a strongly continuous Prabhakar resolvent family S(t), B: U -> X is a bounded linear operator, and f: [0, T] x X -> X is a continuous nonlinear function.

> ***Lemma 2.*** *3 (Mild Solution Representation). If Assumption 2.1 holds, the mild solution of the Prabhakar control system is given by x(t) = S(t) x_0 + int_0^t S(t - s) [B u(s) + f(s, x(s))] ds for all t in [0, T].*

3. Controllability Results

To establish controllability, we introduce the controllability Gramian operator W: U -> X defined by W u = int_0^T S(T - s) B u(s) ds. We assume that the linear operator W has an induced inverse W^{-1} taking values in U/ker(W).

Assumption 3. *1 (Gramian Invertibility and Boundedness). The linear operator W is surjective and there exists M_w > 0 such that ||W^{-1}|| <= M_w. Furthermore, ||S(t)|| <= M_s and ||B|| <= M_b for all t in [0, T].*

Assumption 3. *2 (Lipschitz Continuity of f). The nonlinear function f: [0, T] x X -> X satisfies ||f(t, x_1) - f(t, x_2)|| <= L_f ||x_1 - x_2|| for all x_1, x_2 in X and t in [0, T].*

Theorem 3. *4 (Exact Controllability of Prabhakar Systems). Suppose Assumptions 3.1 and 3.2 hold. If the condition L_f M_s T (1 + M_s M_b M_w T) < 1 is satisfied, then the nonlinear Prabhakar fractional dynamical system is exactly controllable on [0, T].*

Proof. Define an operator Phi: C([0, T], X) -> C([0, T], X) and design the feedback control u(t) = B* S*(T - t) W^{-1} [x_1 - S(T) x_0 - int_0^T S(T - s) f(s, x(s)) ds]. By applying the Banach fixed point theorem on the closed ball B_r, Phi has a unique fixed point x*(t), which satisfies x*(T) = x_1. This completes the proof.

Remark 3. *5. Unlike traditional Caputo systems where kernels are singular power functions, the Prabhakar derivative incorporates exponential memory decay through the Mittag-Leffler kernel, permitting wider stability margins.*